



BUSINESS MATHEMATICS

MAX MARKS: 100
80 (THEORY) +20 (INTERNAL ASSESSMENT)

TIME: 3 HOURS

Unit 1st: Sets, Relations and Functions

10 Marks

Sets and their representation, various types of sets, complement of a set. Algebra of sets (Union, intersection and difference of sets). Demorgan's laws, Cartesian product of sets.

Relations: Various types of relations, Equivalence relation simple examples

Definition of a function and its various types (Into, onto, one-one, many-one, polynomial function, rational, modulus, constant, signum, greatest integer function, composite function).

Unit 2nd: Sequences and Series

10 Marks

Geometric progression, general term sum to n terms, and sum to infinity of a geometric series. Geometric and arithmetic means, Evaluation of $\sum n$, $\sum n^2$, $\sum n^3$

Unit 3rd: Trigonometry

10 Marks

Trigonometric ratios of allied angles (without proof). sum, difference formulae and their applications. solution of trigonometric equations.

Unit 4th: Permutations and combinations

10 Marks

Factorial notation, fundamental principle of counting. Meaning of $P(n,r)$ and $C(n,r)$ and their relations with simple applications.

Unit 5th: Binomial theorem

08 Marks

Binomial theorem for any index. General term, middle term/s of a Binomial Expansion. Applications of binomial expansion.



Unit 6th: Statistics

13 Marks

Measures of dispersion, Mean Deviation from mean and median.

Standard deviation and variance of a grouped and ungrouped data.

Quartile deviation.

Unit 7th: Probability

11 Marks

Random experiment and sample space (set representation). Events and their occurrence. various types of events. Mutually exclusive and Exhaustive events . Axiomatic probability with applications.

Unit 8th: Linear Inequations

08 Marks

Algebraic solution of an inequation in one variable and the representation on a number line. Graphical solution of linear inequations in two variables.

Books Suggested:

1. A Textbook of Mathematics for Class XI published by NCERT, New Delhi

Scheme of Assessment (Theory)

Question paper contains four sections namely

1. **Section – A** contains 10 (Objective Type Question/Multiple Choice Questions) of 1 mark each. $1 \times 10 = 10$ marks
2. **Section- B** contains 10 Very Short Answer Type Questions of 2 marks each. $2 \times 10 = 20$ marks
3. **Section – C** contains 8 Short Answer type questions of 4 marks each. $4 \times 8 = 32$ marks
4. **Section - D** contains 3 Long Answer type questions of 6 marks each. (With Internal Choice) $6 \times 3 = 18$ marks

Note: The paper setter shall incorporate Higher Order Thinking Skills (HOTS) questions apart from knowledge, memory and application based questions.

INTERNAL ASSESSMENT	20 MARKS
Periodic Tests (Best 2 out of 3 tests conducted)	10 Marks
Mathematics Activities	10 Marks